

CS/DS 4342: Machine Learning

Homework 2

Due: November 9th, 2024 11:59pm

Instructions

- **Collaboration:** You must complete this assignment individually; Discussions about the homework to understand the problems are permitted, but sharing problem solutions or code is not allowed.
- **Formatting:** The assignment consists of two programming assignments. You must submit your solutions electronically as a .ipynb file containing the codes and results for both assignments.
- **Citing Sources:** Cite any sources of help (except the course materials) in your PDF.
- **Usage of AI tools:** Responsible and sensible use of generative AI (e.g., ChatGPT) is permitted. However, any use of generative AI must be documented, clearly stating how and for what task it has been used, along with the specific ways it contributed to your work.
- **Submission:** Submit the completed HW2.ipynb file, which should include the code you implemented and the test results for both parts of the assignment.

Objective: This assignment aims to train logistic regression classifiers using mini-batch stochastic gradient descent (mini-batch SGD), as discussed in the lecture. The assignment has two parts: the first part involves implementing logistic regression for binary classification, and the second part focuses on implementing softmax regression (multinomial logistic regression) for multi-class classification.

Part I: Logistic Regression for Binary Classification

In this assignment, you will implement logistic regression for binary classification using stochastic gradient descent (SGD). Mini-batch stochastic gradient descent (mini-batch SGD) is a variant of the stochastic gradient descent (SGD) optimization algorithm that updates the model parameters using *small, randomly selected* subsets (mini-batches) of the training data rather than the entire dataset (gradient descent). You will complete the function prototypes

provided in the `HW2.ipynb` file under the *Logistic Regression* section. Use the provided test code in the same file to validate your implementation, and report the results in your Jupyter Notebook.

Part II: Softmax Regression (Multinomial Logistic Regression)

In this assignment, you will implement softmax regression (multinomial logistic regression) for multi-class classification using mini-batch stochastic gradient descent (SGD). Mini-batch stochastic gradient descent (mini-batch SGD) is a variant of the stochastic gradient descent (SGD) optimization algorithm that updates the model parameters using *small, randomly selected* subsets (mini-batches) of the training data rather than the entire dataset (gradient descent). You will complete the function prototypes provided in the `HW2.ipynb` file under the *Softmax Regression (multinomial logistic regression)* section.

To test your code, you will train your classifier to classify images of clothing articles from the Fashion MNIST dataset. The input to the model will be 28×28 -pixel images (converted into 784-dimensional vectors), and the output will be vectors of 10 probabilities. Test code is provided in the same file. You will run this code to validate your implementation and report the results in your Jupyter Notebook.